

Notes: Grossman and Helpman (QJE, 2002)

Integration versus Outsourcing in Industry Equilibrium

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1 Big picture

- **Question.** Why do some industries organize production inside vertically integrated firms, while others outsource specialized inputs to independent suppliers?
- **Core idea.** The paper develops an industry-equilibrium theory of the make-or-buy decision. A firm's organizational choice depends not only on its own bilateral transaction costs, but also on how many other firms choose to specialize, search, and outsource.
- **Main trade-off.** Vertical integration avoids search frictions and holdup, but integrated firms face higher governance and production costs. Outsourcing allows specialization and potentially lower production costs, but requires costly matching and exposes suppliers to imperfect-contracting problems.
- **Key equilibrium object.** The central comparison is between two demand thresholds, A_I and A_O . A_I is the industry demand level required for vertically integrated firms to break even; A_O is the demand level required for outsourcing pairs to break even.
- **Organizational regimes.** With constant returns to matching, mixed equilibria with both vertically integrated firms and specialized firms are nongeneric. The industry tends to have either pervasive integration or pervasive outsourcing.
- **Stability result.** Vertical integration is stable when $A_I < A_O$. Outsourcing exists and is stable when $A_I > A_O$. Intuitively, the organizational form that can break even at the lower demand level crowds out the other.
- **Comparative statics.** Outsourcing is favored by a large production-cost advantage of specialized suppliers, low fixed costs for specialized firms, and efficient matching/search technology.
- **Market competitiveness.** The effect of final-good substitutability is subtle. When specialized production has a large cost advantage, more elastic final-good demand can favor outsourcing. When the cost advantage is small, the relation may reverse or become nonmonotonic.
- **Market thickness.** With increasing returns to matching, larger industries or larger economies can support outsourcing because thick markets make it easier for buyers and suppliers to find partners.
- **Partial specialization.** In the extended model, inputs need not be perfectly specialized. Suppliers choose how specialized their components should be, balancing higher value in the intended relationship against weaker outside options.
- **Economic takeaway.** Outsourcing is an equilibrium industrial-organization outcome. It depends on specialization benefits, contractual frictions, search technology, market size, and the organizational choices of other firms.

2 Incremental contribution of the paper

- **From bilateral make-or-buy to industry equilibrium.** Earlier incomplete-contracts and transaction-cost theories analyze a single buyer-supplier relationship. This paper embeds those forces in an industry equilibrium with entry, search, pricing, and free entry.

- **Endogenous organizational structure.** Integration and outsourcing are not assumed. They emerge from the interaction of specialized final-good producers, specialized input suppliers, and vertically integrated firms.
- **Search externalities in organization.** The attractiveness of outsourcing depends on the number of potential partners. A specialized final-good firm benefits when more input suppliers enter, and a supplier benefits when more potential customers enter.
- **Equilibrium feedback.** Individual organizational choices change market thickness and product-market competition, which in turn change the profitability of organizational forms.
- **No generic coexistence result.** Under constant returns to matching and homogeneous firms, the paper shows that mixed equilibria are knife-edge. This sharp result clarifies why entire industries may be dominated by one organizational mode.
- **Comparative statics for outsourcing trends.** The model connects outsourcing to improvements in search technology, changes in final-market competition, bargaining power, and market size. This provides a theoretical language for studying rising domestic and international outsourcing.
- **Thick-market mechanism.** With increasing returns to matching, the model explains why outsourcing may be more prevalent in large economies or dense industrial clusters.
- **Endogenous input specificity.** The partial-specialization extension lets suppliers choose whether to make components highly specific or more generic, linking asset specificity, outside options, and bargaining power.
- **General-equilibrium contribution.** The paper offers a tractable framework for studying firm boundaries, prices, product variety, entry, and supplier specialization jointly.

3 Model environment and primitives

3.1 Consumers and differentiated final goods

The economy has J industries. In each industry, firms produce a continuum of differentiated varieties. The representative consumer has CES-style preferences within each industry:

$$u = \sum_{j=1}^J \mu_j \log \left[\int_0^{N_j} y_j(i)^{\alpha_j} di \right]^{1/\alpha_j},$$

where μ_j is the spending share on industry j , N_j is the number of varieties, and $\alpha_j \in (0, 1)$ governs the degree of substitutability among varieties.

Demand for a variety can be written as

$$y_j(i) = A_j p_j(i)^{-1/(1-\alpha_j)},$$

where A_j is an industry demand shifter taken as given by each individual firm.

Interpretation: the parameter α is important because it determines how competitive the product market is. Higher α means varieties are closer substitutes and markups are lower.

3.2 Two ways to organize production

Production of each final-good variety requires a specialized component. The paper compares two organizational forms:

- **Vertical integration.** A single firm designs the final good and produces its own component. It avoids search and holdup, but uses $\lambda > 1$ units of labor per unit of component because of governance costs and reduced specialization.
- **Outsourcing.** A specialized final-good producer searches for a specialized input supplier. If matched, the supplier produces the component and the final-good firm assembles and sells the product. Specialized suppliers can produce the high-quality component with one unit of labor per unit.

Fixed costs differ by organizational type. A vertically integrated firm pays k_v . A specialized final-good producer pays k_s , and a specialized component producer pays k_m . The paper assumes $k_s + k_m \leq k_v$, so specialized production can have lower fixed costs in addition to lower variable costs.

Interpretation: vertical integration economizes on transaction costs, while outsourcing economizes on specialization and governance costs.

3.3 Incomplete contracts and holdup

The quality and detailed characteristics of components are observable to the buyer and supplier but not verifiable by courts. Ex ante contracts cannot fully specify quality-contingent prices and quantities. After the supplier produces the component, the buyer and supplier bargain over the surplus.

If a supplier produces $x(i)$ units for final good i , the joint revenue from the relationship is $p(i)x(i)$. Because the input has no alternative use in the simple model, the joint surplus is divided by bargaining: the input supplier receives share ω and the final-good producer receives share $1 - \omega$.

Interpretation: the supplier anticipates that it will only capture part of the relationship surplus. This generates underproduction relative to the joint-profit-maximizing level and raises the final-good price under outsourcing.

3.4 Search and matching

Let s be the number of specialized final-good producers and m the number of specialized component producers. The number of matches is $n(s, m)$. Under constant returns to matching, the probability that a final-good producer finds a supplier is

$$\eta(r) = n(1, r), \quad r = \frac{m}{s},$$

and the probability that a component producer finds a customer is $\eta(r)/r$.

Interpretation: the two sides of the market are strategic complements. More suppliers make outsourcing more attractive to final producers; more final producers make entry more attractive to suppliers.

3.5 Timing

The game has five stages:

1. firms enter as vertically integrated firms, specialized final-good producers, or specialized input producers;
2. specialized firms search for partners;
3. matched suppliers and integrated firms produce components;
4. matched specialized firms bargain over the surplus;
5. final goods are assembled and sold.

Free entry drives expected profits to zero for any organizational type that enters.

4 Pricing and zero-profit conditions

4.1 Outsourcing prices and profits

A specialized input supplier anticipates capturing only share ω of final revenue. This yields the outsourcing price

$$p_s = \frac{1}{\alpha\omega},$$

which is higher than the price a unitary specialized actor would charge. Output of an outsourced final-good variety is

$$y_s = A(\alpha\omega)^{\alpha/(1-\alpha)}.$$

Expected profit of a specialized final-good producer is

$$\pi_s = \eta(r)(1 - \omega)A(\alpha\omega)^{\alpha/(1-\alpha)} - k_s.$$

Expected profit of a specialized component producer is

$$\pi_m = \frac{\eta(r)}{r}(1 - \alpha)\omega A(\alpha\omega)^{\alpha/(1-\alpha)} - k_m.$$

Interpretation: outsourcing profitability depends on both operating surplus and matching probabilities. The holdup problem appears through the distorted price and the reduced output induced by $\omega < 1$.

4.2 Vertical integration prices and profits

An integrated firm faces marginal cost λ . Its monopoly price under CES demand is

$$p_v = \frac{\lambda}{\alpha},$$

and its output is

$$y_v = A\left(\frac{\alpha}{\lambda}\right)^{\alpha/(1-\alpha)}.$$

Expected profit of an integrated firm is

$$\pi_v = (1 - \alpha)A \left(\frac{\alpha}{\lambda} \right)^{\alpha/(1-\alpha)} - k_v.$$

Interpretation: vertical integration avoids the supplier holdup distortion but has a higher marginal cost λ and a potentially higher fixed governance cost.

4.3 The demand thresholds A_O and A_I

For outsourcing to be viable, both specialized final-good and component producers must break even. The zero-profit conditions pin down the ratio of specialized entrants:

$$r_0 = \frac{\omega(1 - \alpha)}{1 - \omega} \frac{k_s}{k_m},$$

and the industry demand level required for outsourcing:

$$A_O = \frac{(\alpha\omega)^{-\alpha/(1-\alpha)} k_m}{\omega(1 - \alpha)} \frac{r_0}{\eta(r_0)}.$$

For vertical integration, the zero-profit condition defines

$$A_I = \frac{(\lambda/\alpha)^{\alpha/(1-\alpha)} k_v}{1 - \alpha}.$$

The entire equilibrium comparison is organized around A_I/A_O .

Interpretation: lower A_O makes outsourcing easier to sustain; lower A_I makes vertical integration easier to sustain.

5 Equilibrium logic and comparative statics

5.1 Mixed equilibria are nongeneric

A mixed equilibrium would require positive entry by all three types: integrated firms, specialized final-good producers, and specialized component producers. This requires both outsourcing and integration to break even at the same industry demand level. Generically this fails because the zero-profit condition for outsourcing pins down A_O , while the zero-profit condition for integration pins down A_I .

$$A_O = A_I$$

is a knife-edge condition. Hence, with homogeneous firms and constant returns to matching, industries do not normally contain both organizational forms.

Interpretation: this result is not saying that real-world coexistence never occurs. It says that in this stylized homogeneous model, one organizational form tends to dominate in equilibrium.

5.2 Vertical integration equilibrium

There is always a possible equilibrium in which all entrants are vertically integrated, because a lone specialized entrant would find no partner and lose its entry cost. However, this equilibrium is stable only when

$$A_I < A_O.$$

If a small mass of specialized firms enters and outsourcing can break even at a lower demand level, the vertically integrated equilibrium is not stable.

Interpretation: vertical integration is stable when integrated firms can survive at lower demand than outsourcing pairs.

5.3 Outsourcing equilibrium

An outsourcing equilibrium requires positive entry by both specialized final-good producers and specialized component producers. The equilibrium numbers are

$$s_O = \frac{(1 - \omega)\mu L}{k_s}, \quad m_O = \frac{(1 - \alpha)\omega\mu L}{k_m}.$$

Pervasive outsourcing exists if and only if

$$A_I \geq A_O,$$

and is stable when $A_I > A_O$.

Interpretation: outsourcing is stable when specialized firms can break even at a lower demand level than integrated firms.

5.4 Search efficiency, cost advantage, and fixed costs

The key ratio is

$$\frac{A_I}{A_O} = \omega(\lambda\omega)^{\alpha/(1-\alpha)} \frac{\eta(r_0)}{r_0} \frac{k_v}{k_m}.$$

Outsourcing becomes more likely when this ratio rises. This happens when:

- λ is high, meaning integrated firms have a larger variable-cost disadvantage;
- k_v is high relative to k_s and k_m , meaning vertical integration has a large governance fixed cost;
- matching is efficient, so $\eta(r_0)/r_0$ is large;
- search frictions are low, so specialized firms expect to find partners more often.

Interpretation: improvements in search and matching - for example through better communication technology, B2B platforms, or industrial clustering - can raise the viability of outsourcing.

5.5 Elasticity of substitution

The effect of α is subtle. A higher α means varieties are closer substitutes and final-good demand is more elastic.

If $\lambda\omega > 1$, specialized outsourcing has a sufficiently strong price/cost advantage. Then increasing α raises A_I/A_O and makes outsourcing more viable.

If $\lambda\omega < 1$, the direct price effect favors vertical integration as demand becomes more elastic. But a higher α also changes the relative entry of specialized producers and can improve the matching probability of component suppliers. As a result, the relationship between α and outsourcing can be decreasing or hump-shaped.

Interpretation: more competition does not mechanically favor outsourcing. Its effect depends on whether specialized suppliers' production advantage is large enough to overcome the holdup-induced pricing distortion.

5.6 Bargaining power

The bargaining share ω affects outsourcing through three channels:

- a higher ω gives component suppliers a larger share of revenue;
- a higher ω reduces the holdup distortion by giving suppliers stronger incentives to produce;
- a higher ω attracts more suppliers relative to final-good producers, lowering the match probability for each supplier.

These effects imply an inverted-U relation: outsourcing is least viable when suppliers have very little or very much bargaining power, and most viable at intermediate levels of ω .

Interpretation: efficient outsourcing requires balanced incentives. If suppliers get too little surplus, they underproduce. If suppliers get too much surplus, too many suppliers enter and each supplier faces a low match probability.

6 Extensions: market thickness and input specialization

6.1 Increasing returns to matching

Under constant returns to matching, industry size does not determine whether outsourcing is viable. With increasing returns to matching, this changes. If more firms on both sides of the market make it easier for each firm to find a partner, large markets can support outsourcing more easily.

The required demand level for outsourcing becomes

$$A_O = \frac{(\alpha\omega)^{-\alpha/(1-\alpha)} k_m}{\omega(1-\alpha)} \frac{r_0 s}{n(s, r_0 s)}.$$

With increasing returns, a larger s lowers A_O , because the matching technology works better in thicker markets.

Interpretation: this provides a theory of why outsourcing and firm specialization may be more prevalent in dense industrial clusters, large cities, and large economies.

6.2 Multiple stable equilibria

Increasing returns to matching can generate multiple stable equilibria. One stable equilibrium may have pervasive vertical integration, while another has pervasive outsourcing.

Interpretation: history and expectations can matter. A market with few specialized firms may remain integrated because outsourcing is initially unattractive; a thick specialized market may sustain itself once established.

6.3 Partial specialization and secondary markets

The simple model assumes that a component made for one final producer has zero value elsewhere. The extension relaxes this. Each final good has an ideal component, but nonideal components can still be used at an adjustment cost.

The input supplier chooses the degree of specialization ρ_i :

- a highly specialized component is valuable to the intended customer;
- a more generic component has better resale value in the secondary market and improves the supplier's outside option.

Interpretation: suppliers choose partial specialization. They do not always make perfectly tailored inputs, because genericity improves bargaining power.

6.4 Specificity and outsourcing

A natural intuition is that higher specificity should favor integration because specialized assets intensify holdup. The paper shows that this need not be true. Higher specificity reduces outside value, but it also changes bargaining positions and equilibrium output. Under some conditions, outsourcing can be viable even when input specifications are quite important.

Interpretation: asset specificity does not mechanically imply integration. Its equilibrium effect depends on how it changes output, bargaining power, and entry.

7 Identification and theoretical design

7.1 What the model identifies

This is a theory paper, not an empirical identification exercise. The model identifies mechanisms through which firm boundaries are shaped in general equilibrium:

- search costs and matching probabilities;
- incomplete contracts and holdup;

- specialization advantages;
- bargaining power;
- product-market competition;
- market thickness and increasing returns to matching.

Interpretation: the purpose is to explain why outsourcing can be an industry-wide equilibrium rather than a collection of independent bilateral choices.

7.2 Why equilibrium matters

A bilateral make-or-buy model treats the set of outside suppliers as fixed. Grossman and Helpman endogenize that set. The probability of finding a supplier depends on how many suppliers enter, and their entry depends on how many buyers want to outsource.

Interpretation: outsourcing can become self-reinforcing. Once many specialized firms exist, outsourcing becomes easier; when few exist, integration may persist.

7.3 Limits of the framework

- Firms are homogeneous within type, which helps generate sharp no-coexistence results.
- Search is highly stylized and summarized by a matching function.
- Bargaining shares are exogenous.
- Contracting incompleteness is assumed rather than derived from a microfoundation.
- The model abstracts from repeated relationships, reputation, relational contracts, and rich contract menus.

Interpretation: these simplifications make the equilibrium logic transparent. The contribution is not empirical realism in every detail, but a tractable framework for studying industrial organization in equilibrium.

8 Figures

8.1 Figures I and II: equilibrium curves and stability

Read:

- Figure I shows the case $A_I > A_O$. The outsourcing zero-profit curve lies outside the vertical-integration zero-profit curve. The outsourcing equilibrium E_O is stable, while the vertical-integration point E_V is unstable.
- Figure II shows the case $A_I < A_O$. The vertical-integration curve lies outside the outsourcing curve. The integrated equilibrium E_V is stable.
- The arrows show the entry-exit dynamics. Firms enter when expected profits are positive and exit when expected profits are negative.

- The figures illustrate why the comparison of A_I and A_O determines the stable organizational form.

Economic message: The organizational form that can break even at the lower industry demand level is the stable one. Outsourcing dominates when specialized firms can survive at lower demand; integration dominates when integrated firms can survive at lower demand.

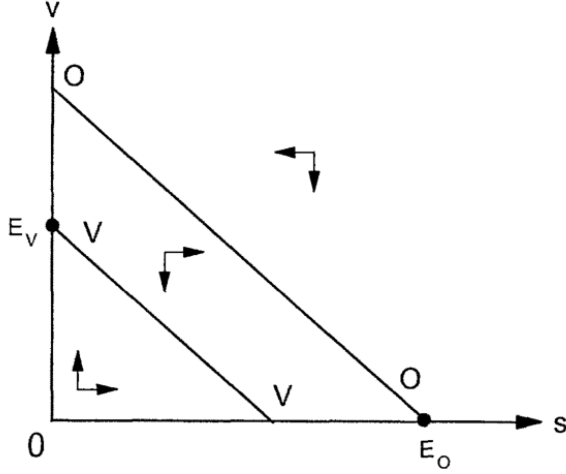


FIGURE I
Equilibrium Curves: $A_I > A_O$

(a) Figure I: $A_I > A_O$

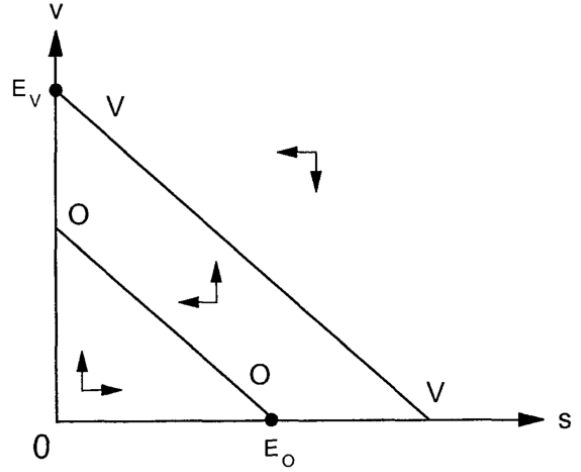


FIGURE II
Equilibrium Curves: $A_I < A_O$

(b) Figure II: $A_I < A_O$

8.2 Figures III and IV: elasticity of substitution and industry organization

Read:

- Figure III considers $\lambda\omega > 1$. In this case, outsourcing becomes more viable as α rises. A stable outsourcing equilibrium exists for sufficiently high substitutability, $\alpha > \alpha_1$.
- Figure IV considers $\lambda\omega < 1$. Panel a shows a monotone decreasing relationship between A_I/A_O and α , so outsourcing is viable only at low α .
- Figure IV, Panel b shows a hump-shaped relationship. Outsourcing is viable at intermediate values of α .
- The different shapes reflect two opposing forces: product-market competition changes operating profits, while changes in entry composition affect match probabilities.

Economic message: More competitive final-good markets can favor or hurt outsourcing. The sign depends on the strength of specialized suppliers' cost advantage relative to the holdup distortion.

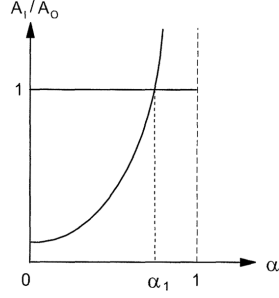


FIGURE III

Industry Organization for $\lambda\omega > 1$: The Role of the Elasticity of Substitution

(a) Figure III: $\lambda\omega > 1$

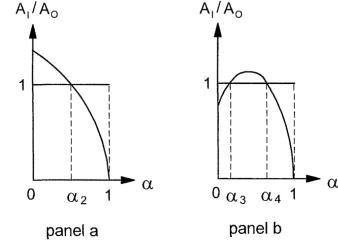


FIGURE IV

Industry Organization for $\lambda\omega < 1$: The Role of the Elasticity of Substitution

(b) Figure IV: $\lambda\omega < 1$

8.3 Figures V and VI: bargaining power and increasing returns to matching

Read:

- Figure V shows that A_I/A_O is hump-shaped in supplier bargaining power ω . Outsourcing is viable only for intermediate bargaining power.
- When ω is too low, suppliers have weak incentives and outsourcing is unattractive. When ω is too high, too many suppliers enter relative to final-good producers, reducing each supplier's match probability.
- Figure VI introduces increasing returns to matching. The outsourcing zero-profit curve can become nonlinear and can intersect the integration curve.
- With increasing returns, both pervasive integration and pervasive outsourcing can be locally stable. The outcome can depend on market thickness and initial conditions.

Economic message: Outsourcing requires balanced surplus sharing and a sufficiently thick specialized market. This creates a natural theory of industrial clusters and path dependence.

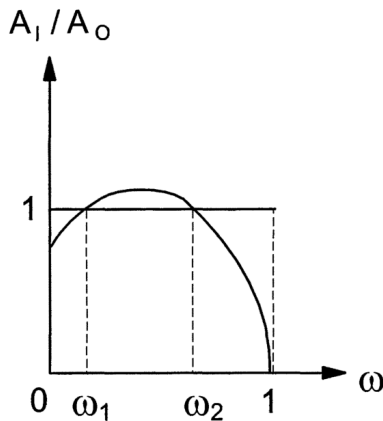


FIGURE V

Industry Organization: The Role of Bargaining Power

(a) Figure V: bargaining power

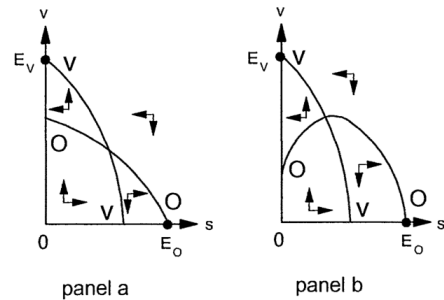


FIGURE VI

Increasing Returns to Matching: Multiple Equilibria

(b) Figure VI: increasing returns to matching

8.4 Figure VII: input characteristics and partial specialization

Read:

- Figure VII represents input characteristics in a two-dimensional space.
- Point i is the ideal input for final good i , while point h is the ideal input for another final good h .
- Point s_i is a component designed partly for final good i . The distance from s_i to i measures how imperfectly specialized the input is for its intended use.
- The parameter ρ_i measures the degree of specialization toward i , while θ_{ih} captures similarity between the ideal inputs for final goods i and h .
- A more specialized input is more valuable in its intended use but less valuable in secondary markets.

Economic message: Suppliers choose partial specificity to trade off relationship value against outside-option value. This extends the model beyond the extreme assumption that specialized inputs are worthless outside their intended relationship.

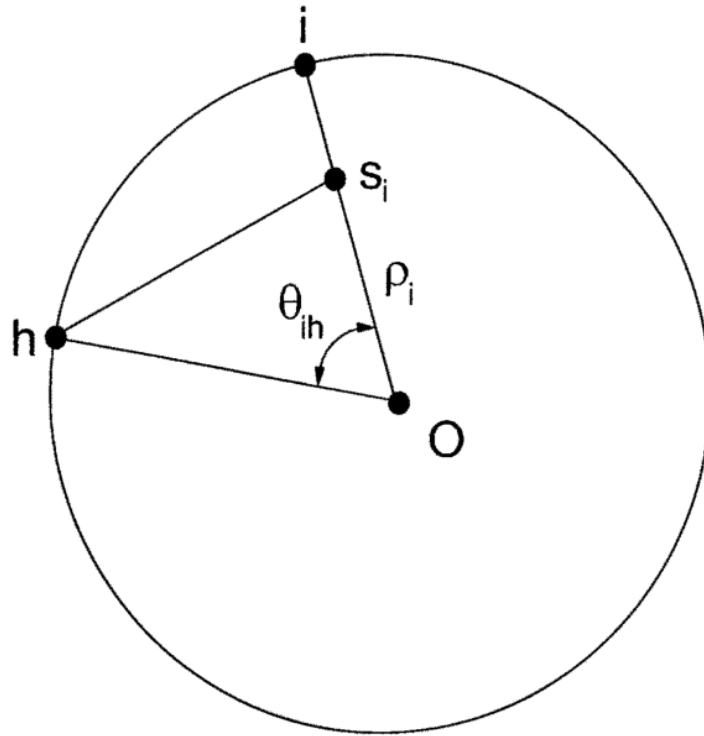


FIGURE VII
Input Characteristics

Figure VII: input characteristics

9 Short summary

- The paper builds a general-equilibrium model of firm boundaries.
- Final goods can be produced either by vertically integrated firms or by outsourcing pairs of specialized firms.
- Integration avoids search and holdup but has higher governance and production costs.
- Outsourcing benefits from specialization but suffers from search frictions and incomplete-contracting distortions.
- With constant returns to matching, mixed equilibria are nongeneric.
- The key comparison is A_I versus A_O : integration is stable if $A_I < A_O$, and outsourcing is stable if $A_I > A_O$.
- Outsourcing is favored by specialized suppliers' cost advantage, lower fixed costs, and more efficient matching.
- Product-market competition has ambiguous effects because it changes both operating margins and matching probabilities.
- Intermediate supplier bargaining power has an inverted-U effect on outsourcing viability.
- With increasing returns to matching, large and thick markets can support outsourcing, and multiple stable equilibria can arise.
- In the partial-specialization extension, suppliers choose how specific their inputs should be, balancing intended-use value against outside options.

10 Conclusion

10.1 Core conclusion

Grossman and Helpman show that firm boundaries are equilibrium objects. The make-or-buy decision cannot be fully understood by looking only at a single bilateral relationship, because the profitability of outsourcing depends on how many potential suppliers and customers enter the market.

10.2 Economic intuition

The central intuition is a trade-off. Integrated firms solve the holdup and search problems internally, but they lose the cost advantages of specialization and pay higher governance costs. Specialized firms can be more efficient, but their success depends on finding partners and managing incomplete-contracting problems. In equilibrium, the industry selects the organizational form that survives more easily under entry and competition.

10.3 Incremental contribution revisited

The paper’s main contribution is to move incomplete-contracts theories of firm boundaries into a general-equilibrium industrial-organization setting. It shows how outsourcing depends on matching technology, market size, competition, bargaining power, and the organization choices of other firms. This helps explain cross-industry, cross-region, and over-time differences in outsourcing.

10.4 Implications

The model implies that outsourcing can rise when search improves, when markets become thicker, or when specialized suppliers gain a large production-cost advantage. It also implies path dependence: once a specialized supplier base forms, outsourcing may become self-sustaining. This provides a framework for thinking about industrial clusters, business-to-business markets, and international outsourcing.

10.5 Limitations

The model is deliberately stylized. It uses homogeneous firms, simple matching, exogenous bargaining power, and a stark incomplete-contracting friction. It abstracts from repeated relationships, reputation, detailed contract design, managerial heterogeneity, and empirical measurement. These assumptions sharpen the equilibrium logic but limit direct quantitative application.

10.6 Takeaway

The enduring takeaway is that outsourcing is not just a firm’s private procurement choice. It is an industry-equilibrium phenomenon shaped by market thickness, contracting frictions, and the endogenous availability of specialized partners.